SUNWAY UNIVERSITY · SCHOOL OF MATHEMATICAL SCIENCES & COMPUTING

Autonomous Wildfire Risk Assessment & UAV Surveillance

Mamdani fuzzy inference driving a four-probe UAV inside an NVIDIA Isaac Sim digital twin

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Isaac Sim OpenUSD stage — sensor bridge

NVIDIA Isaac Sim 6.2.0 · OpenUSD procedural stage · 4 inputs · 16 sets · 10 rules · 0.07 ms median inference · p99 0.23 ms

01 Why a threshold fails

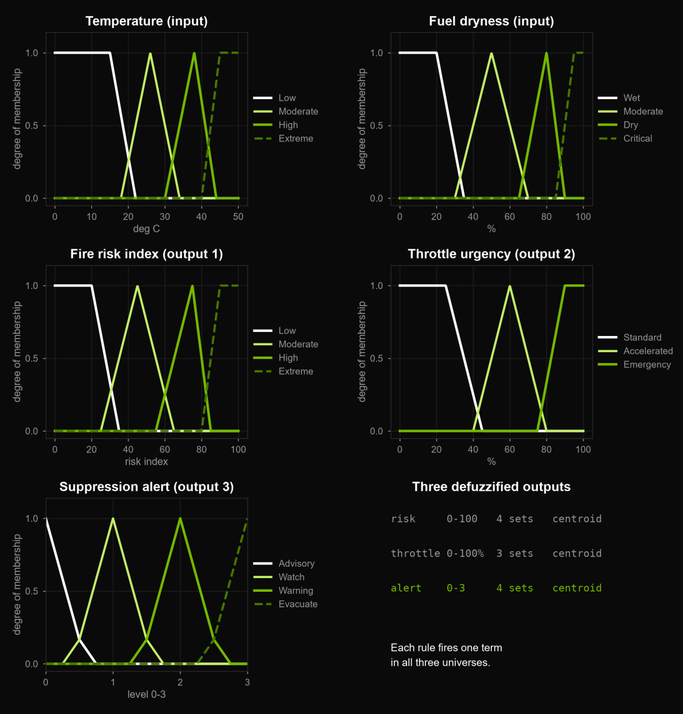
A single cut-off cannot separate **26 °C at 15 % humidity with 65 km/h wind** from **38 °C under 85 % humidity**. The cooler case is the dangerous one: the controller scores it **FRI 71.4** and the hot, humid case **FRI 14.0**, while a temperature-only rule ranks them the other way round. Dry wind at moderate temperature ignites faster than a hot, humid afternoon.

BEHAVIOUR	THRESHOLD	MAMDANI
Decision surface	step at 40 °C	**continuous**
Humidity / wind	ignored	**fused**
Sensor noise	chatters at the edge	**averaged**
Output	alarm on / off	**throttle 20–90 % + alert 0–3**

02 Linguistic variables

Four inputs, four sets each — 16 in total. Trapezoids close the outer sets, so a reading past the sensor range saturates instead of running off the universe.

VAR	UNIVERSE	SETS
T	0–50 °C	low · moderate · high · extreme
RH	0–100 %	very low · low · moderate · high
WS	0–100 km/h	calm · moderate · high · severe
FD	0–100 %	wet · moderate · dry · critical



Membership functions — temperature and fuel dryness shown; humidity and wind use the same four-set template. Outputs: fire risk index (4 sets), throttle urgency (3), suppression alert (4 sets on 0–3).

03 Rule base

4 inputs × 4 sets = 256 antecedent combinations, curated down to the 10 that are physically meaningful. **any** means the rule does not constrain that variable; min for AND, max for aggregation. Every rule fires all three consequents at once.

RULE	T	RH	WS	FD	RISK	THR.	ALERT
R1	any	High	any	Wet	Low	Standard	Advisory
R2	Low	Moderate	any	any	Low	Standard	Advisory
R3	Moderate	any	any	Moderate	Moderate	Standard	Watch
R4	Moderate	any	High	Dry	High	Accelerated	Warning
R5	High	Low	any	any	High	Accelerated	Warning
R6	High	any	Severe	any	Extreme	Emergency	Evacuate
R7	Extreme	Very Low	any	any	Extreme	Emergency	Evacuate
R8	any	any	High	Critical	Extreme	Emergency	Evacuate
R9	Extreme	any	any	Critical	Extreme	Emergency	Evacuate
R10	any	High	Calm	any	Low	Standard	Advisory

Firing strength α = min(μ of each antecedent); aggregated output μagg(y) = max over rules. The alert consequent is a separate fuzzy inference on [0, 3], so it is not a threshold applied to the risk index.

04 Defuzzification

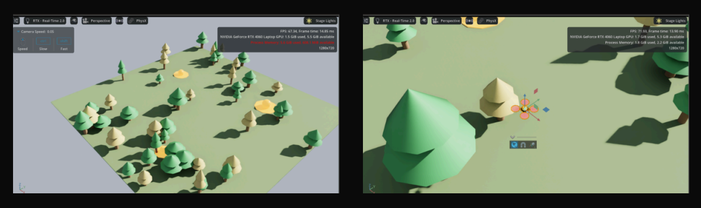
$$y^* = \int y \cdot \mu_{agg}(y)\, dy \,/\, \int \mu_{agg}(y)\, dy$$

Fire risk index under each method, four scenarios. The three methods do not separate on speed; they separate on continuity. Last column is the centroid-to-MOM gap.

SCENARIO	COG	BISECTOR	MOM	GAP
Benign Clear Day	17.5	17.5	17.5	0.0
Dry Spring Heat	55.6	55.6	56.2	0.6
High Hazard Drought	70.3	70.3	70.6	0.3
Critical Wildfire	92.2	92.5	95.1	**2.9**

MOM returns the mean of the aggregated set's maxima, so crossing a boundary jumps the output between peaks — a torque step on a 250 g airframe. Centroid integrates the whole area, so its output is **continuous in the input**.

05 Isaac Sim digital twin



65 m OpenUSD stage, 47 procedural trees, 3 active hotspots, chase and thermal cameras. Probe readings are taken from the stage at every keyframe, so the controller is in the loop.

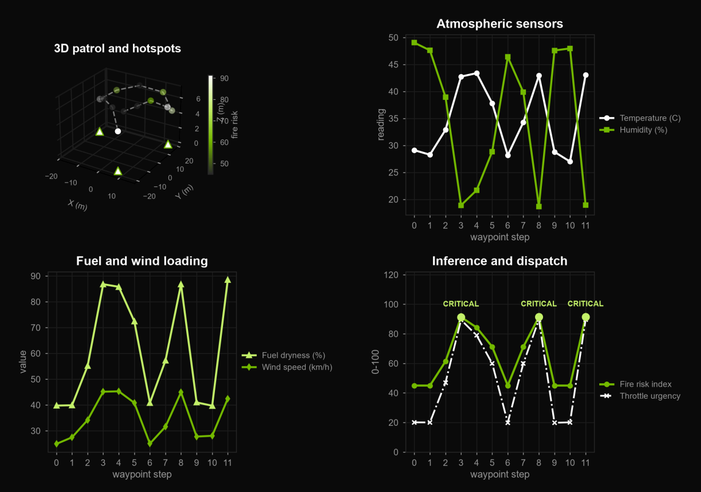
65 × 65 m — procedural OpenUSD stage

3 — hotspots, 44.5 °C cores

14 — keyframes, frames 0–620

0.07 ms — median loop, p99 0.23 ms

06 Flight results



WP	TEMP	RH	FRI	THR.	ALERT
WP 00	29.1 °C	49.0 %	45.0	20.1 %	WATCH
WP 03	42.8 °C	18.9 %	91.4	89.5 %	**CRITICAL**
WP 06	28.2 °C	46.4 %	45.0	19.9 %	WATCH
WP 08	42.9 °C	18.7 %	91.4	89.6 %	**CRITICAL**

Twelve waypoints, seed 0, no retuning. 91.4 is a centroid output, not a hardcoded 90; recovery returns to 45.0 so nothing latches.

45.0 — FRI on open ground, 5 of 12 waypoints

91.4–91.5 — FRI at the three hotspot passes

2.83 — alert value at the hotspot passes

1.00 — alert value on open ground

The alert track does not simply mirror the risk track: waypoint 04 reads FRI 84.3 against 79.0 % throttle but only alert 2.49, because fuel dryness falls to 85.8 % while wind stays at 45.3 km/h. A single threshold on the risk index cannot express that.

07 Conclusion

Four inputs, ten auditable rules and a closed loop inside Isaac Sim; **25 unit tests pass**. **Limitation:** simulation only, rule base expert-tuned, edge latency projected rather than measured. **Next:** ANFIS tuning, multi-UAV mesh, PX4 and Matrice 300 field trials.

Impact. Reporting a hotspot while it is still small is what makes containment cheap: the earlier the coordinates reach the crew, the less water, airtime and exposure the response costs (SDG 13.1, SDG 15.2).

github.com/123haiku/csc3034-fuzzy-wildfire-sim
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